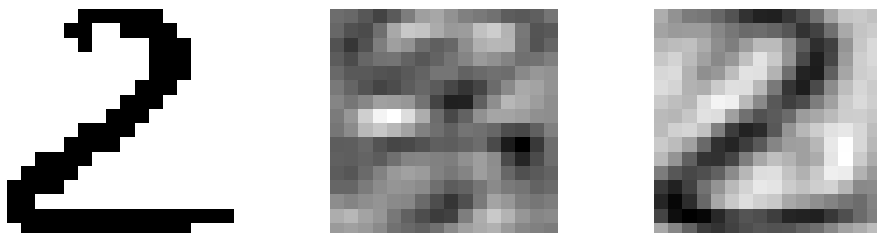


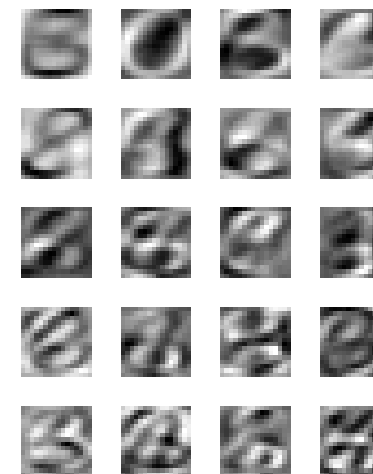
Principal Component Analysis (PCA)

1.6 -1.1 -1.6 2.1 -0.52 2.8 0.72 0.7 -0.68 -0.41 -1.4 -1.5 -0.54 -0.62 1.3 -1.4 -0.27 0.74 0.77 -1



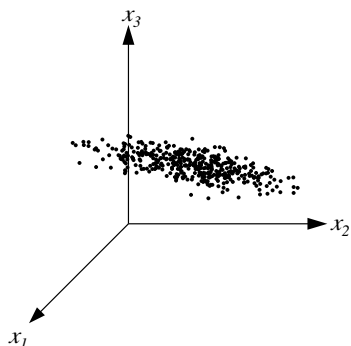
Covariance matrices, dimensionality reduction, PCA, Duality

1. **Covariance Matrices**
2. Principal Component Analysis
3. Duality



Spread of Data

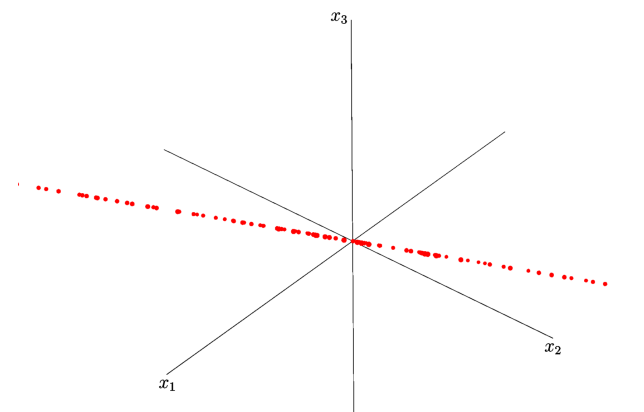
- Often data varies significantly in only some directions



- Reduce dimensions by projecting onto low dimensional subspace with maximum variation

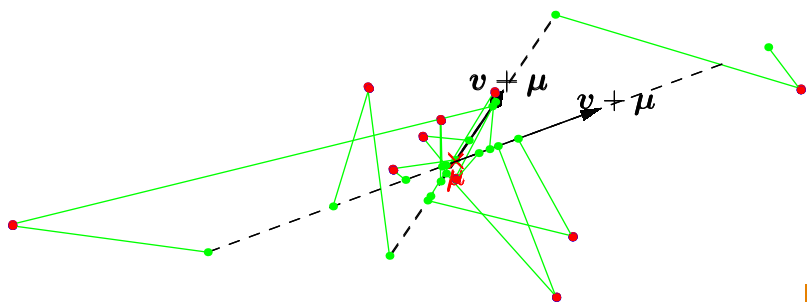
Looking is not Enough

Can't spot low dimensional data by looking at numbers



Dimensionality Reduction

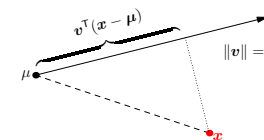
- Often helpful to consider only directions where data varies significantly
- Want to find directions along which data has its greatest variation



Direction of Maximum Variation

- Look for the vector v with $\|v\|^2 = 1$ to maximise

$$\sigma^2 = \frac{1}{m-1} \sum_{i=1}^m (v^\top (x_i - \mu))^2$$



- This is a constrained optimisation problem
- Solve by maximising Lagrangian

$$\mathcal{L} = \frac{1}{m-1} \sum_{k=1}^m (v^\top (x_k - \mu))^2 - \lambda (\|v\|^2 - 1)$$

- λ is a Lagrange multiplier

Direction of Maximum Variation

- Expanding the Lagrangian

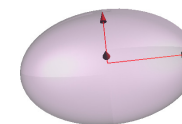
$$\begin{aligned} \mathcal{L} &= \frac{1}{m-1} \sum_{k=1}^m (v^\top (x_k - \mu))^2 - \lambda (\|v\|^2 - 1) \\ &= \frac{1}{m-1} \sum_{k=1}^m (v^\top (x_k - \mu) (x_k - \mu)^\top v) - \lambda (\|v\|^2 - 1) \\ &= v^\top \left(\frac{1}{m-1} \sum_{k=1}^m (x_k - \mu) (x_k - \mu)^\top \right) v - \lambda (\|v\|^2 - 1) \\ &= v^\top C v - \lambda (v^\top v - 1) \end{aligned}$$

- Extrema of the Lagrangian

$$\nabla \mathcal{L} = 2(Cv - \lambda v) = 0 \Rightarrow Cv = \lambda v$$

Direction of Maximum Variation

- The eigenvectors are directions that are extrema of the variance



- The variance in direction v is equal to

$$\begin{aligned} \sigma^2 &= \frac{1}{m-1} \sum_{i=1}^m (v^\top (x_i - \mu))^2 \\ &= v^\top C v = \lambda v^\top v = \lambda \end{aligned}$$

- The variance is maximised by the eigenvector with the maximum eigenvalue

Covariance Matrix

- The **covariance matrix** is defined as

$$\mathbf{C} = \frac{1}{m-1} \sum_{k=1}^m (\mathbf{x}_k - \boldsymbol{\mu})(\mathbf{x}_k - \boldsymbol{\mu})^\top$$

- The components C_{ij} measure how the i^{th} and j^{th} components co-vary

$$C_{ij} = \frac{1}{m-1} \sum_{k=1}^m (x_{ik} - \mu_i)(x_{jk} - \mu_j)$$

- C.f. covariance of random variables

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

Matrix Form

- The covariance matrix is

$$\mathbf{C} = \frac{1}{m-1} \sum_{k=1}^m (\mathbf{x}_k - \boldsymbol{\mu})(\mathbf{x}_k - \boldsymbol{\mu})^\top$$

- Define the matrix

$$\mathbf{X} = \frac{1}{\sqrt{m-1}} (\mathbf{x}_1 - \boldsymbol{\mu}, \mathbf{x}_2 - \boldsymbol{\mu}, \dots, \mathbf{x}_m - \boldsymbol{\mu})$$

- We can write the covariance matrix as

$$\mathbf{C} = \mathbf{X}\mathbf{X}^\top$$

Outer Product

- Remember that the outer-product of two vectors is defined as

$$\mathbf{x}\mathbf{y}^\top = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \begin{pmatrix} y_1 & y_2 & \dots & y_n \end{pmatrix} = \begin{pmatrix} x_1 y_1 & x_1 y_2 & \dots & x_1 y_n \\ x_2 y_1 & x_2 y_2 & \dots & x_2 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_n y_1 & x_n y_2 & \dots & x_n y_n \end{pmatrix}$$

- C.f. Inner product

$$\mathbf{x}^\top \mathbf{y} = \begin{pmatrix} x_1 & x_2 & \dots & x_n \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

Properties of Covariance Matrix

- The **quadratic form** of a vector and matrix is defined as

$$\mathbf{v}^\top \mathbf{M} \mathbf{v}$$

- The quadratic form of a covariance matrix is non-negative for any vector

$$\mathbf{v}^\top \mathbf{C} \mathbf{v} = \mathbf{v}^\top \mathbf{X} \mathbf{X}^\top \mathbf{v} = \mathbf{u}^\top \mathbf{u} = \|\mathbf{u}\|^2 \geq 0$$

where $\mathbf{u} = \mathbf{X}^\top \mathbf{v}$

- Matrices with non-negative quadratic forms are known as **positive semi-definite**

Eigenvalue Decomposition

- The eigenvectors of $\mathbf{C}$ with the largest eigenvalues are known as the **principal components**
- The eigenvalues are all greater than or equal to zero
- Recall an eigenvector $\mathbf{v}$ satisfies the equation

$$\mathbf{C}\mathbf{v} = \lambda\mathbf{v}$$

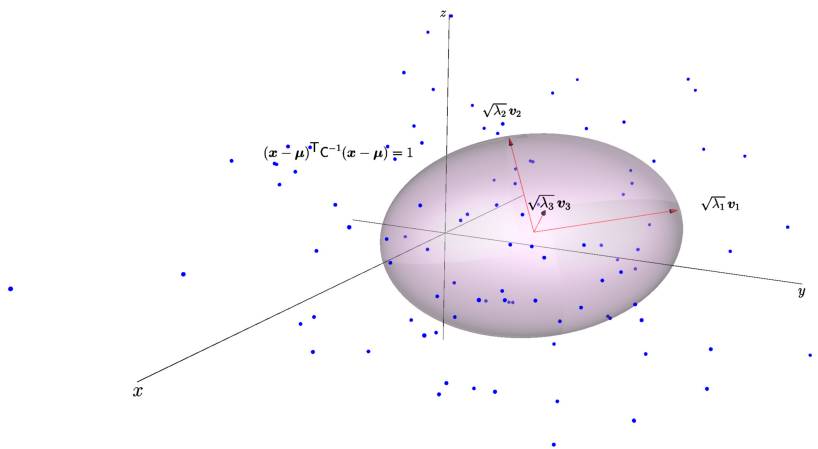
- Multiplying both sides by $\mathbf{v}^T$

$$\mathbf{v}^T\mathbf{C}\mathbf{v} = \lambda\mathbf{v}^T\mathbf{v} = \lambda\|\mathbf{v}\|^2$$

but $\mathbf{v}^T\mathbf{C}\mathbf{v} \geq 0$ and $\|\mathbf{v}\|^2 > 0$ so

$$\lambda = \frac{\mathbf{v}^T\mathbf{C}\mathbf{v}}{\|\mathbf{v}\|^2} \geq 0$$

Ellipsoid and Eigen Space



Surface Defined by Matrix

- The set of vectors $\mathbf{x}$ such that

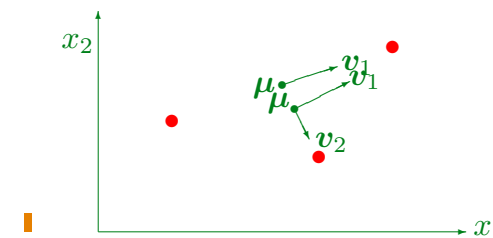
$$\mathbf{x}^T\mathbf{C}^{-1}\mathbf{x} = 1$$

defines a surface

- The surface is an ellipsoid, $\mathcal{E}$
- The eigenvectors point in the direction of the principal axes of the ellipsoid
- The radii of the principal axes are equal to the square root of the eigenvalues

Spanning Input Space

- A covariance matrix will have a zero eigenvalue only if there is no variation in the direction of the corresponding eigenvector
- A covariance matrix will have zero eigenvalues if the number of patterns are less than or equal to the number of dimensions
- A covariance matrix formed from $p + 1$ patterns that do not all lie in a hyperplane (i.e. the p differences $\mathbf{x}_k - \mathbf{x}_1$ are linearly independent) will have no zero eigenvalues

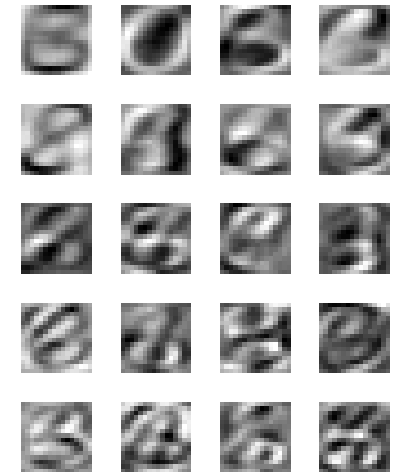


- Matrices with no zero eigenvalues are called **full rank** matrices (as opposed to rank deficient)
- Full rank matrices are invertible, rank deficient matrices are singular and non-invertible
- Full rank covariance matrices have only positive eigenvalues and are said to be **positive definite**
- We would expect that when $m > p$ the covariance matrix will be positive definite unless there are some symmetries that linearly constrain the patterns

Principal Component Analysis

- PCA occurs as follows
 - ★ Construct the covariance matrix
 - ★ Find the eigenvalues and eigenvectors
 - ★ Keep the eigenvectors with the largest eigenvalues (principal components)
 - ★ Project the inputs into the space spanned by the principal components
- We then use the projected inputs as inputs to our learning machine

1. Covariance Matrices
2. **Principal Component Analysis**
3. Duality



Projection Matrix

- To project the inputs construct the projection matrix

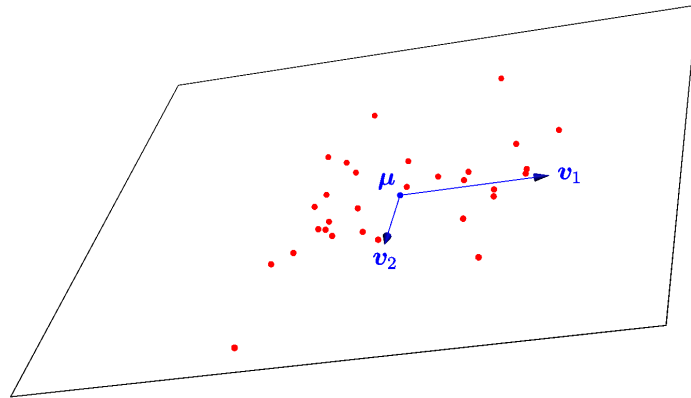
$$\mathbf{P} = \begin{pmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_k^T \end{pmatrix}$$

- $k < p$ is the number of principal components we keep
- Given a p -dimensional input pattern $\mathbf{x}$ we can construct a k -dimensional representation $\mathbf{z}$

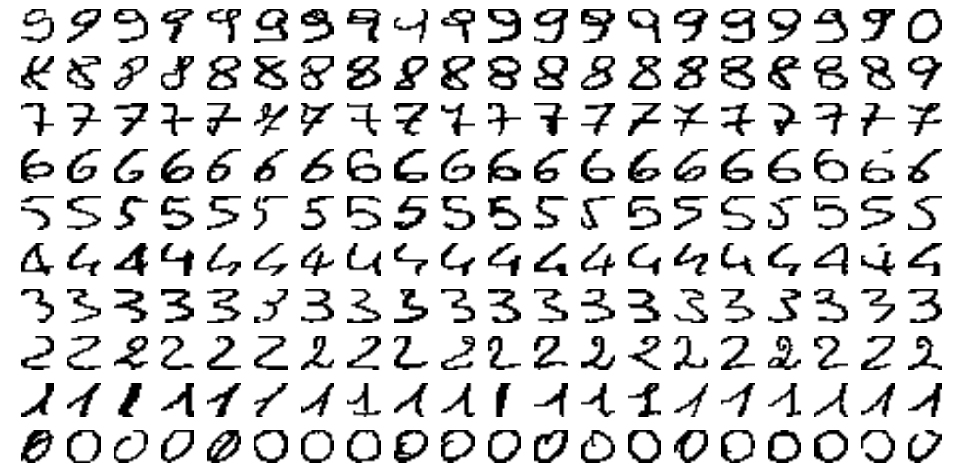
$$\mathbf{z} = \mathbf{P}(\mathbf{x} - \boldsymbol{\mu})$$

- Use $\mathbf{z}$ as our new inputs

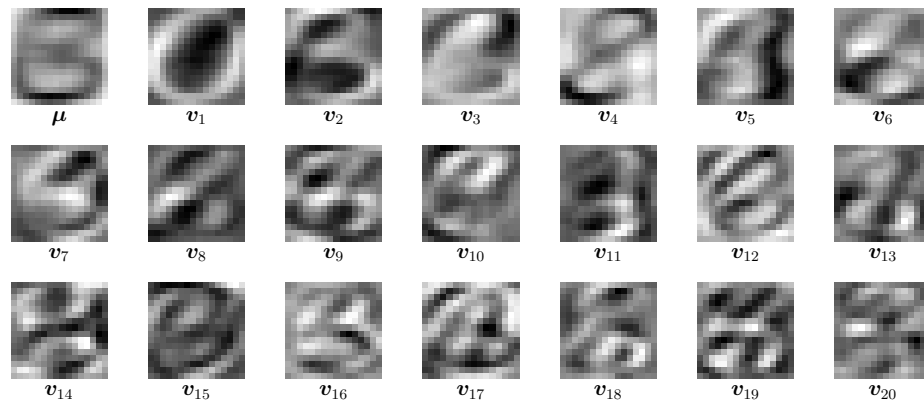
Subspace Projection



Hand Written Digits



Eigenvectors



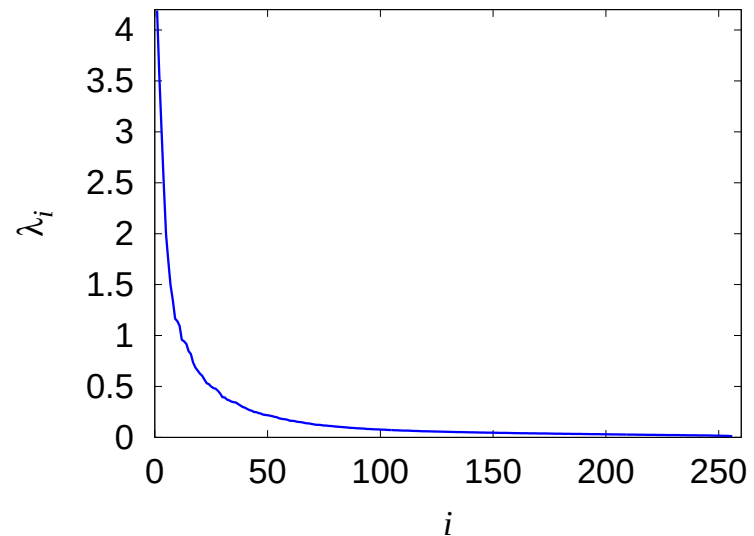
Reconstruction

- Projecting into a subspace of eigenvectors can be seen as approximating the inputs by

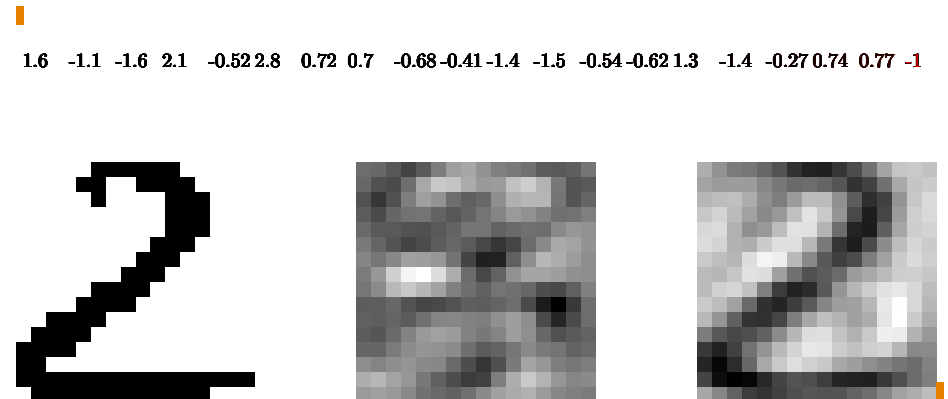
$$\hat{x}_i = \mu + \sum_{j=1}^k z_j^i v_j, \quad z_j^i = v_j^T (x_i - \mu), \quad \|v_j\| = 1$$

- Principal component analysis projects the data into the subspace of dimension k with the minimal approximation error $\mathbb{E}[\|\hat{x}_i - x_i\|^2]$
- The loss of “energy” (or squared error) is equal to the sum of the eigenvalues in the directions that are ignored

Eigenvalues for Digits

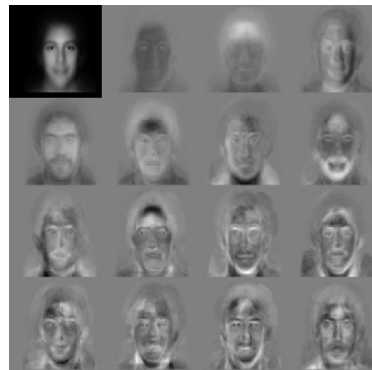


Reconstruction from Eigenvectors



Outline

1. Covariance Matrices
2. Principal Component Analysis
3. **Duality**



PCA for Images

- An image often contains around $p = 256 \times 256 = 64k$ pixels
- In standard PCA we would create an $p \times p$ matrix with over 4×10^9 elements
- This is intractable
- m images span at most a $m - 1$ dimensional subspace
- Usually this subspace will be much smaller than the space of all images $m \ll p$

Dual Matrix

- The covariance $\mathbf{C} = \mathbf{X}\mathbf{X}^\top$ is a $p \times p$ matrix
- Consider the $m \times m$ matrix $\mathbf{D} = \mathbf{X}^\top\mathbf{X}$
- Suppose $\mathbf{v}$ is an eigenvector of $\mathbf{D}$

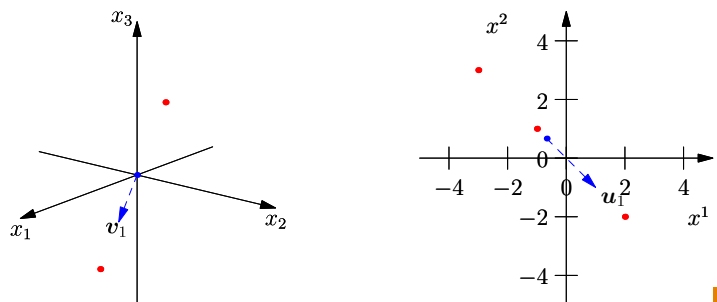
$$\begin{aligned}\mathbf{D}\mathbf{v} &= \lambda\mathbf{v} \\ \mathbf{X}^\top\mathbf{X}\mathbf{v} &= \lambda\mathbf{v} \\ \mathbf{X}\mathbf{X}^\top\mathbf{X}\mathbf{v} &= \lambda\mathbf{X}\mathbf{v} \\ \mathbf{C}\mathbf{X}\mathbf{v} &= \lambda\mathbf{X}\mathbf{v} \Rightarrow \mathbf{C}\mathbf{u} = \lambda\mathbf{u}\end{aligned}$$

- $\mathbf{u} = \mathbf{X}\mathbf{v}$ (and $\mathbf{v} \propto \mathbf{X}^\top\mathbf{u}$)

What Does a Subspace Look Like?

- Consider $\mathbf{y}^1 = \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix}$, $\mathbf{y}^2 = \begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix}$ with mean $\boldsymbol{\mu} = \begin{pmatrix} 5 \\ 5 \\ 3 \end{pmatrix}$
- Subtracting the mean $\mathbf{x}^i = \mathbf{y}^i - \boldsymbol{\mu}$ we can construct matrix

$$\mathbf{X} = \begin{pmatrix} x_1^1 & x_1^2 \\ x_2^1 & x_2^2 \\ x_3^1 & x_3^2 \end{pmatrix} = \begin{pmatrix} -3 & 3 \\ -1 & 1 \\ 2 & -2 \end{pmatrix}$$



Dual Matrix

- Matrices $\mathbf{C} = \mathbf{X}\mathbf{X}^\top$ and $\mathbf{D} = \mathbf{X}^\top\mathbf{X}$ have the same non-zero eigenvalues
- Can use the dual $m \times m$ matrix $\mathbf{D}$ to find eigenvalues and eigenvectors of $\mathbf{C}$
- Note that $\mathbf{D} = \mathbf{X}^\top\mathbf{X}$ has components $D_{kl} \propto (\mathbf{x}_k - \boldsymbol{\mu})^\top(\mathbf{x}_l - \boldsymbol{\mu})$
- Takes $O(p \times m \times m)$ time to construct $\mathbf{D}$
- We work in a “dual space” which is the space spanned by the examples

Summary

- PCA allows us to reduce the dimensionality of the inputs
- We project the inputs into a sub-space where the data varies the most
- We can work in either the original space ($\mathbf{X}\mathbf{X}^\top$) or the dual space ($\mathbf{X}^\top\mathbf{X}$)
- When we have many more features than examples (i.e. $p \gg m$) then it is more efficient working in the dual space
- We will see examples of dual spaces again when we look at SVMs